Math 242, S20 **Name:** Answer

Quiz 5

<u>**Instructions:**</u> **There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.**

Problem 1. (5 points) Solve the following initial value problem for $P(t)$:

$$\frac{dP}{dt} = 8P - 2P^2, \qquad P(0) = 5.$$

and find $\lim_{t\to+\infty} P(t)$.

$$\frac{dP}{dt} = 2P(4-P)$$

$$\frac{1}{P(4-P)}\,dP = 2dt$$

$$\int \frac{1}{P(4-P)}\,dP = \int 2\,dt.$$

$$\int \frac{1}{4}\left(\frac{1}{P} + \frac{1}{4-P}\right)dP = \int 2\,dt.$$

$$\int \frac{1}{P} + \frac{1}{4-P}\,dP = \int 8\,dt.$$

$$\ln|P| - \ln|4-P| = 8t + C$$

$$\ln\left|\frac{P}{4-P}\right| = 8t + C$$

$$\Rightarrow \left|\frac{P}{4-P}\right| = e^{8t+C} \Rightarrow \frac{P}{4-P} = Be^{8t} \ (B\neq 0)$$

$$P(0) = 5 \Rightarrow \frac{5}{4-5} = B \Rightarrow B = -5$$

thus $\frac{P}{4-P} = -5e^{8t}$

$$P = (4-P)(-5e^{8t})$$

$$P = -20e^{8t} + 5Pe^{8t}$$

$$(1-5e^{8t})P = -20e^{8t}$$

$$P = \frac{-20e^{8t}}{1-5e^{8t}} = \frac{20}{5-e^{-8t}}$$

$$\lim_{t\to+\infty} P(t) = \lim_{t\to+\infty} \frac{20}{5-e^{-8t}} = \frac{20}{5} = 4$$

Problem 2. (5 points) Given the following autonomous differential equation:

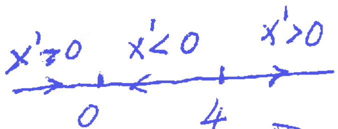
$$\frac{dx}{dt} = 2x^2 - 8x.$$

Find all critical points of the given differential equation and then determine whether it is stable or unstable for each of them.

$$f(x) = 2x^2 - 8x = 2x(x-4)$$

$$f(x) = 0 \Rightarrow 2x(x-4) = 0 \Rightarrow x = 0 \text{ or } x = 4$$

Thus it has two critical points: $x=0$, $x=4$



$x' = f(x)$

i) if $x<0$, then $x'>0$

ii) if $0<x<4$, then $x'<0$ $\Rightarrow$ $x=0$ stable, $x=4$ unstable.

iii) if $x>4$, then $x'>0$